

NETWORKING AND ROUTING PROTOCOL

ET 8315

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Mr. Cuthbert John Karawa

Lecture 3

CONFIGURING A ROUTER

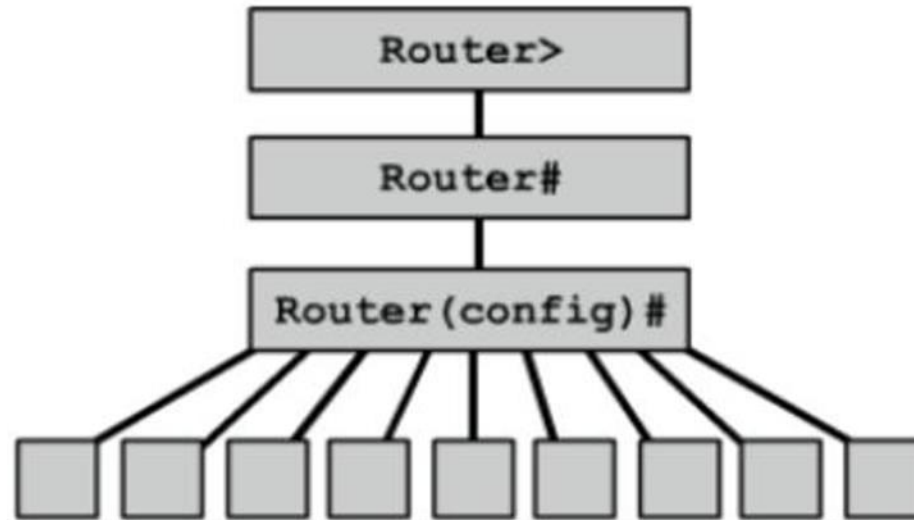
**College of Information and Communication Technology
(CoICT)**

Department Of Electronics And Telecommunications Engineering.

Basic Router Configuration

- When **configuring a router**, certain basic tasks are performed, including the following:
 - ✓ Naming the router
 - ✓ Setting passwords
 - ✓ Configuring interfaces
 - ✓ Configuring a banner
 - ✓ Saving changes on a router
 - ✓ Verifying basic configuration and router operations

CLI Command Modes



Configuration Mode	Prompt
Interface	Router(config-if)#
Subinterface	Router(config-subif)#
Controller	Router(config-controller)#
Map-list	Router(config-map-list)#
Map-class	Router(config-map-class)#
Line	Router(config-line)#
Router	Router(config-router)#
IPX-router	Router(config-ipx-router)#
Route-map	Router(config-route-map)#

Basic Router Configuration

- The **first prompt** is at user mode:
 - ✓ Router>
- User mode will allow you to view **the state of the router** but will **not allow you to modify its** configuration.
- Don't confuse “**user mode**” with “**users of the network.**”
- “User mode” is intended for the **network technicians, operators, and engineers** who have the responsibility to configure network devices.

Basic Router Configuration

- The **enable** command is used to enter **privileged EXEC** mode. This mode allows the user to **make configuration changes on the router**.
- The router prompt will change from a > to a # in this mode:
 - ✓ Router> enable
 - ✓ Router#

Host Name and Passwords

- Naming the router
 - ✓ Syntax: Router(config)# **hostname** *name*
- First, enter global configuration mode:
 - ✓ Router# **config t**
- Next, apply a unique host name to the router:
 - ✓ Router(config)# **hostname dmitri**
 - ✓ dmitri(config)#

Host Name and Passwords

- Now, **configure a password** that is to be **used to enter** privileged EXEC mode. We will use the password **class**.
 - ✓ dmitri(config)# enable secret class
- Next, configure the **console** and **Telnet** lines with the password **must**.
- The **login** command enables password checking on the line.
- If you do not enter the **login command** on the console line, the user will be granted access to the line without entering a password.

Host Name and Passwords

- **The console commands follow:**
 - ✓ **dmitri**(config)# line console 0
 - ✓ **dmitri**(config-line)# password must
 - ✓ **dmitri**(config-line)# login
- **The Telnet lines use similar commands:**
 - ✓ **dmitri**(config)# line vty 0 4
 - ✓ **dmitri**(config-line)# password must
 - ✓ **dmitri**(config-line)# login

Host Name and Passwords

- By default, **line passwords** are stored in **clear-text** in configuration files. To ensure these passwords are encrypted in all configuration files:
 - ✓ dmitri(config)# **service password-encryption**
- Aux is used to set the **user-mode password** for the **auxiliary port**. This is typically used for configuring a modem on the router but can be used as a console as well.
- **Console** is used to set a **console user-mode password**
- **Vty** is used to set a **Telnet password on the router**. If the password is not set, then Telnet cannot be used by default.

Configuring a Banner

```
dmitri con0 is now available
```

```
Press RETURN to get started.
```

```
ONGEA NA CUTHBERT KAMA UANATAKA KUINGIA
```

```
User Access Verification
```

```
Password:
```

```
dmitri>enable
```

```
Password:
```

```
dmitri#confi
```

```
dmitri#configure ter
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
dmitri(config)#banner motd #ONGEA NA ADMIN KUPATA ACCESS#
```

```
dmitri(config)#
```

- You can set a **banner** on a router so that when either a user logs into the router or an administrator telnets into the router for example, a **banner will give them information you want them to have.**
- Another reason for having banner is to add a security notice to users dialing into your internetwork.

Configuring a Banner

- From global configuration mode, configure the message-of-the-day (MOTD) banner. A **delimiting character** such as a # is used at the beginning and at the end of the message. The delimiter allows you to configure a **multiline banner** as shown here:
 - ✓ dmitri(config)# **banner motd** # **WARNING!!**
Unauthorized Access Prohibited!!#
- Configuring an appropriate banner is part of a **good security plan**.
- At a minimum, a banner should warn against unauthorized access.

Router Interface Configuration

- You will now configure the **individual router interfaces** with **IP addresses** and **other information**.
- First, enter **interface configuration mode** by specifying the **interface type** and **number**.
- Next, configure the **IP address** and **subnet mask**:
- **Serial interface configuration:**
 - ✓ dmitri# show interfaces **OR**
 - ✓ dmitri# show ip interface brief
 - ✓ dmitri(config)# interface GigabitEthernet0/0/0
 - ✓ dmitri(config-if)# ip address **192.168.2.1 255.255.255.0**

Router Interface Configuration

- It is good practice to configure a **description** on each interface to **help document the network information**. The description text is **limited** to **240 characters**.
- On production networks, a description can be helpful in **troubleshooting** by providing information about the type of network the interface is connected to and whether any other routers are on that network.
 - ✓ `dmitri(config-if)# description Link to R2`

Router Interface Configuration

- Set clock rate if a **DCE cable** is connected (skip this step if a DTE cable is connected):
 - ✓ `dmitri(config-if)# clock rate 64000`
- Serial interfaces require a clock signal to **control the timing of the communications**. DCE devices provide the clock.
- By default, Cisco routers are DTE devices but they can be configured as DCE devices.

Router Interface Configuration

- After configuring the **IP address** and **description**, the **interface must be activated** with the **no shutdown** command.
- This is similar to powering on the interface. The interface must also be connected to another device (a hub, a switch, another router, and so on) for the physical layer to be active.
 - ✓ dmitri(config-if)# **no shutdown**

Router Interface Configuration

Serial interface configuration:

```
dmitri(config)# interface Serial0/0/0
```

```
dmitri(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
dmitri(config-if)# description Link to R2
```

```
dmitri(config-if)# no shutdown
```

Fast Ethernet interface configuration:

```
dmitri(config)# interface FastEthernet0/0
```

```
dmitri(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
dmitri(config-if)# description dmitri LAN
```

```
dmitri(config-if)# no shutdown
```


Verifying Basic Router Configuration

- All the previous basic router configuration commands entered were immediately **stored in the running configuration** file of dmitri.
- The running-config file is stored in **RAM** and is the configuration file used by IOS.
- Verify the commands entered by displaying the running configuration with the show running-config command
 - ✓ dmitri# **show running-config**

dmitri# show running-config

```
dmitri#show running-config
Building configuration...

Current configuration : 821 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname dmitri
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCil
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
```

```
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
  description link to router cuth
  ip address 192.168.2.1 255.255.255.0
  duplex auto
  speed auto
!
interface GigabitEthernet0/0/1
  no ip address
  duplex auto
  speed auto
```

Verifying Basic Router Configuration

- Now that the basic configuration commands have been entered, it is important to save the **running-config file to nonvolatile memory**, the router's **NVRAM**.
- In case of a power outage or an accidental reload, the router will be able to boot with the current configuration.
- After the router's configuration has been completed and tested, it is important to save the running-config file to the startup-config file as the permanent configuration file:
 - ✓ dmitri# **copy running-config startup-config**

Verifying Basic Router Configuration

- The **show running-config** command
 - ✓ Displays the current running configuration that is stored in **RAM**.
- The show startup-config command,
 - ✓ Displays the startup configuration file stored in **NVRAM**.
- This is the configuration that the router will use on the **next reboot**.
- This **configuration does not change** unless the current running configuration is saved to NVRAM with the **copy running-config startup-config** command.

```
dmitri# show startup-config
```

```
dmitri#show startup-config
Using 821 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname dmitri
!
!
!
enable secret 5 $l$mERr$9cTjUIEqNGurQiFU.ZeCi
!
!
!
!
!
ip cef
no ipv6 cef
!
!
```

```
!
spanning-tree mode pvst
!
!
!
!
!
!
!
interface GigabitEthernet0/0/0
description link to router cuth
ip address 192.168.2.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
no ip address
duplex auto
speed auto
shutdown
!
!--More--
```

Verifying Basic Router Configuration

- When comparing the output from the **show running-config** command and the **show startup-config** command, notice that the startup configuration and the running configuration are **identical**.
- They are identical because the running configuration has **not changed since** the last time it was saved.
- Also notice that the show startup-config command displays **how many bytes** of NVRAM the saved configuration is using: 821 bytes.





NEXT LECTURE

**ROUTING AND ROUTING
PROTOCOLS**